

JECRC UNIVERSITY
School of Engineering
B. Tech; I Semester (Common to all)
Subject: Mathematics I (DMA024A)
Tutorial Sheet - 2

Time: 2 hrs.

Max. Marks: 64

Instructions:

1. Attempt all the questions in sequence.
2. Illustrate your answers with suitable examples and diagrams, wherever necessary.
3. Write the relevant question number before writing the answer.

CO1: To apply differential and integral calculus to notions of curvature and to improper integrals. Apart from some other applications, they will have a basic understanding of Beta and Gamma functions.

Part A

(5×2=10)

- A1. [CO1] Define the Definite Integral and its types. Give one example for each.
- A2. [CO1] Define the Beta and Gamma functions.
- A3. [CO1] Evaluate $\Gamma(\frac{5}{2})$ and $\Gamma(\frac{11}{2})$.
- A4. [CO1] Evaluate $\int_0^{\pi/2} \sin^3 x \cos^4 x \, dx$.
- A5. [CO1] Using the property $\Gamma(n)\Gamma(1-n) = \frac{\pi}{\sin(n\pi)}$, evaluate $\Gamma(-\frac{1}{2})$ and $\Gamma(-\frac{3}{2})$.

Part B

(3×7=21)

- B1. [CO1] Evaluate: $\int_0^{\infty} e^{-x^{(1/n)}} \, dx$.
- B2. [CO1] Evaluate: $\int_0^{\pi/2} \tan^n x \, dx$.
- B3. [CO1] Evaluate: $\int_0^1 x^{(n-1)} (\log \frac{1}{x})^{(m-1)} \, dx$.

Part C

(3×11=33)

- C1. [CO1] Evaluate: $\int_0^{\infty} \frac{x^{(m-1)}}{(1+x)^{(m+n)}} \, dx$ and show that it equals $B(m,n)$.
- C2. [CO1] Evaluate: $\int_0^2 x (8-x^3)^{(1/3)} \, dx$.
- C3. [CO1] Show that: $\int_a^b (x-a)^{(m-1)} (b-x)^{(n-1)} \, dx = (b-a)^{(m+n-1)} B(m,n)$.

Answers

Part A

A3: $\frac{3}{4}\sqrt{\pi}$ and $\frac{945}{32}\sqrt{\pi}$. A4: $\frac{2}{35}$. A5: $-2\sqrt{\pi}$ and $\frac{4}{3}\sqrt{\pi}$.

Part B

B1: $\Gamma(n+1)$. B2: $\frac{\pi}{2} \sec(\frac{n\pi}{2})$. B3: $\frac{\Gamma(m)}{n^m}$.

Part C

C2: $\frac{(16\pi)}{(9\sqrt{3})}$.